

Emo-TTA

*Improving Test-Time Adaptation of Audio-Language Models
for Speech Emotion Recognition*

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Adapting at test time

Why this paper matters



Test-time adaptation with no gradients and no batch

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test sample at a time —
no buffering, no batch

0

backpropagation, prompt tuning,
or weight updates

+7.9

average accuracy points over
zero-shot CLAP-HTSAT

10 / 12

dataset–backbone cells where
it is the best method

● The claim

CLAP classifies an utterance by comparing its audio embedding to text prompts for each emotion. Under domain shift those text prototypes sit in the wrong place. So treat them as the initial means of a Gaussian mixture, and re-estimate the means, covariance and priors online from the test stream by expectation-maximisation — using CLAP's own predictions as soft labels.

● The inversion relative to Tent

Tent minimises entropy by gradient descent on normalisation parameters. Emo-TTA uses entropy as a confidence weight — an utterance whose prediction is uncertain contributes less to the statistics — and updates in closed form.

Same signal, opposite role: a loss in one, a reliability score in the other.

1

Why speech emotion is different

Three requirements prior methods do not meet



Why TTA for SER is harder than for ASR



The paper's diagnosis

● The shifts are different in kind

Compared to speech recognition, emotion exhibits greater speaker variability, emotional ambiguity, and weaker linguistic structure.

Real deployment faces shifts in speakers, languages, devices, ambient noise — and, distinctively, inconsistent label taxonomies across corpora.

That last one has no analogue in image corruption benchmarks.

● And the constraints are tighter

Existing TTA methods rely on test batches, model updates, or prompt tuning — making them unsuitable when access to training sources or buffered inputs is severely restricted.

Emotional speech is exactly the kind of data where buffering is a privacy problem, and where utterances arrive one at a time in a conversation.

● Connect to the whole series

Every corpus session documented the taxonomy problem: IEMOCAP's nine categories merged to four, MELD's seven, CREMA-D's six, RAVDESS's eight. A model trained on one label space meets another at test time.

And the CLEP-DG session established how far cross-corpus emotion performance falls. This paper attacks the same gap from the inference side rather than the training side.

Three requirements



And how prior methods score against them

Method family	Training-free	Lightweight	Models the test distribution	Needs
CoOp / CoCoOp — prompt learning	No	No	No	A few labelled target examples
Treff-Adapter — few-shot adapter	No	No	No	Labelled target data and training
TPT — test-time prompt tuning	No	Yes	No	Gradients and augmented views
MTA, ZERO — training-free TTA	Yes	Yes	No	A batch of samples
Emo-TTA	Yes	Yes	Yes	One sample at a time

● The claimed gap

The paper's novelty claim is the conjunction: to the best of our knowledge, we are the first to simultaneously satisfy all three — test-time distribution estimation, lightweight adaptation, and training-free inference.

The third column is the one that is actually new. Training-free TTA already existed; modelling the structure of the test stream did not.

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The method

Gaussian discriminant analysis, estimated online



The starting point: CLAP as a classifier



And why it breaks under shift

● Zero-shot classification

CLAP embeds the audio with $f(\cdot)$ and each class prompt with $g(\cdot)$, then softmaxes the cosine similarities:

$$P(y = i \mid a) \propto \exp(\cos(f(a), g(t_i)))$$

Prompts take the form "This is a [EMOTION] sound." The text encoder supplies the decision boundary.

● Why it degrades

CLAP exhibits performance degradation under acoustic domain shifts — speaker, channel, noise.

The text prototypes stay where they are; the audio embeddings move. The decision boundary is fixed while the data it has to separate has shifted.

● The reframing that makes the method possible

A softmax over cosine similarity to class prototypes is already a generative classifier in disguise — assume class-conditional Gaussians with shared covariance and the posterior takes exactly this softmax form.

So CLAP's text prototypes can be read as initial class means μ_y , with the covariance set to the identity. That reading is what lets you update them with a hundred-year-old algorithm.

Gaussian discriminant analysis



The model behind the update

● The assumption

CLAP audio embeddings for each emotion class follow a multivariate normal with class-specific means μ_y and a shared covariance Σ .

$p(F_t | y) = N(F_t | \mu_y, \Sigma)$, where F_t is the embedding of the t -th utterance.

● The prediction

By Bayes' rule, and taking logs:

$$y^* = \operatorname{argmax}_y [\log P(y) - \frac{1}{2} (F_t - \mu_y)^T \Sigma^{-1} (F_t - \mu_y)]$$

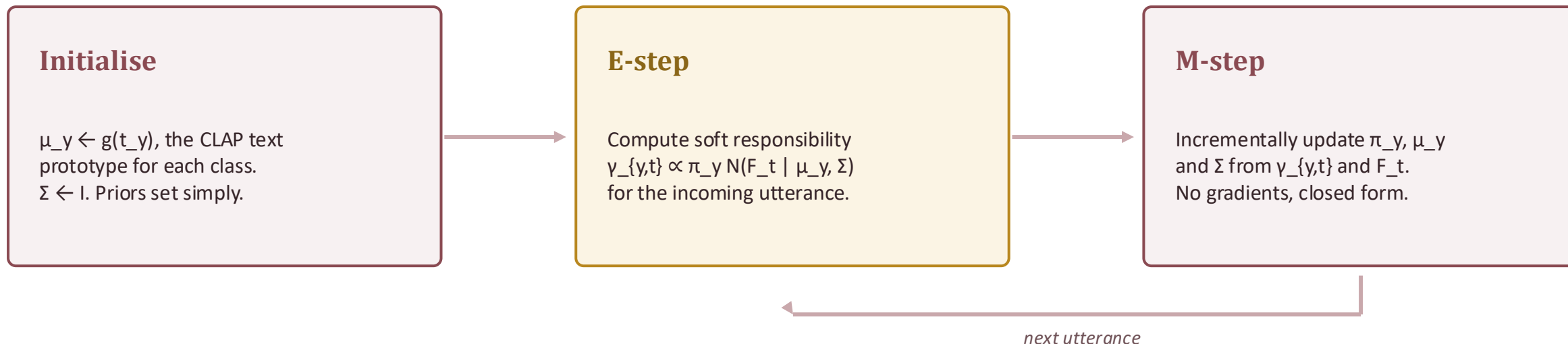
A Mahalanobis distance adjusted for feature correlation and scale, plus a class prior.

- The shared-covariance assumption is what makes the decision boundaries linear and keeps the parameter count manageable.
- Σ is a $d \times d$ matrix where d is CLAP's embedding dimension — so this is not a trivial object to estimate from a few thousand unlabelled utterances.
- The class prior $P(y)$ is also estimated, which means the method adapts to the target corpus's label distribution as well as its acoustics.
- That last point matters here more than in vision: emotion corpora have wildly different class balances, as every corpus session established.

The EM loop



Initialise from text, then update from audio



● What is and is not stored

The method keeps sufficient statistics — means, a covariance, priors and effective counts — not past test samples. That is what supports the privacy claim and the no-buffering claim.

But it does carry state across the stream. Each prediction depends on every utterance seen before it, so this is sample-wise processing, not sample-wise independence.

Entropy as a weight, not a loss



The stabilising mechanism

● The problem it solves

Early in the stream the statistics are still close to their text initialisation, and under distribution shift CLAP's predictions may deviate from emotion semantics.

An early confident-but-wrong responsibility would drag a class mean into the wrong region, and every subsequent update would inherit the error.

● The mechanism

Compute the self-entropy $H(a_t)$ of CLAP's predicted distribution for the utterance, then weight its contribution by

$$w(h) = e^{(-\beta h)}, \text{ with } \beta = 4.5.$$

Confident utterances count fully; uncertain ones barely move the statistics.

● The contrast with Tent in one line

Tent treats low entropy as the goal and takes a gradient step toward it. Emo-TTA treats low entropy as evidence of reliability and uses it to decide how much to trust a sample.

The second use is much safer. Tent's failure on MNIST-to-SVHN was confident-but-wrong predictions being amplified; here confident-but-wrong predictions are still trusted, but nothing pushes the model to become more confident than it already is.

The final prediction



Fusing the frozen classifier with the fitted one

$$\text{logits}_y = \text{T}_y^T \text{F} + \alpha (\text{w}_y^T \text{F} + \text{b}_y)$$

CLAP zero-shot term

generative term, from the fitted Gaussian

- $\text{T}_y = g(\text{t}_y)$ is the CLAP text prototype and $\text{F} = f(\text{a}_t)$ the audio embedding, so the first term is the unmodified zero-shot score.
- $\text{w}_y = \Sigma^{-1} \mu_y$ and $\text{b}_y = \log P(y) - \frac{1}{2} \mu_y^T \Sigma^{-1} \mu_y$ are the standard linear-discriminant weight and bias derived from the fitted Gaussian.
- $\alpha = 0.2$ sets how much the adapted statistics are allowed to move the prediction. The zero-shot term keeps its full weight.
- So the frozen model is never overridden — it is nudged. If the estimation goes wrong, the zero-shot score is still the dominant term.

This is a safety mechanism as much as a fusion rule, and it is a sensible one. It also means the headline gains are achieved while the original classifier still carries most of the decision.

3

Results

Six out-of-domain corpora, two CLAP backbones



Setup



Six corpora, all evaluated out of domain

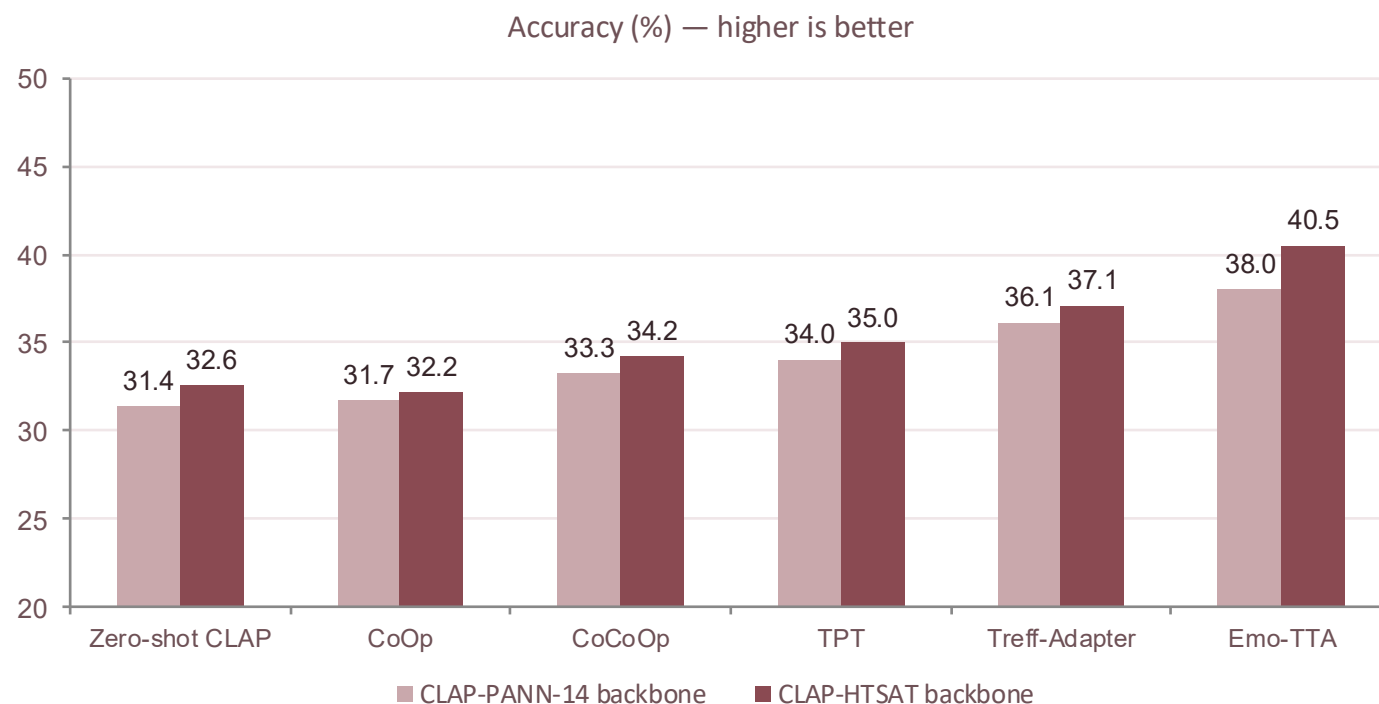
Corpus	Content	Classes	Protocol
IEMOCAP	Dyadic scripted and improvised dialogue, 10 speakers	4	5-fold leave-one-session-out, excited merged into happy
MELD	Multi-party dialogue from Friends	4 (mapped)	Official train/dev/test split
RAVDESS	2.5k studio utterances, about 5 s each	8	5-fold stratified cross-validation
TESS	Acted, consistent recording conditions	4 (mapped)	Random 10-fold cross-validation
SAVEE	Acted, consistent recording conditions	4 (mapped)	Random 10-fold cross-validation
CREMA-D	7k utterances, about 4.5 s each	6	5-fold cross-validation, original classes

- All audio resampled to 16 kHz and standardised to 5 seconds by truncation or zero-padding; prompts are "This is a [EMOTION] sound."
- Two CLAP backbones: PANN-14 and HTS-AT. $\alpha = 0.2$, $\beta = 4.5$, batch size 1, no backpropagation.
- Accuracy is top-1, averaged over three runs with random seeds.
- Every corpus here has appeared in this series. Four of them we have read the original papers for.

Main results



Average accuracy across the six corpora



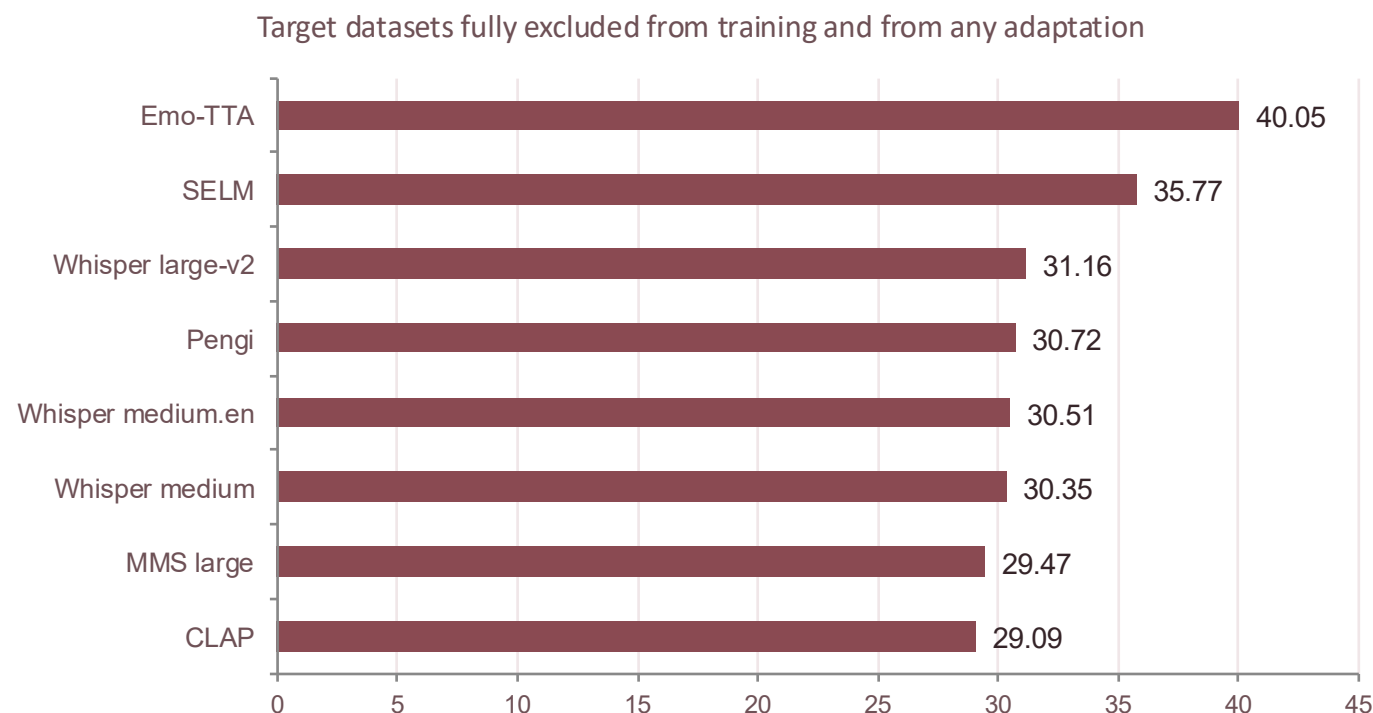
- Emo-TTA leads on both backbones: +6.65 over zero-shot on PANN-14 and +7.90 on HTS-AT.
- It beats Treff-Adapter by 1.91 and 3.37 — and Treff-Adapter uses labelled target data and training.
- Against the two training-free batch methods on HTS-AT, MTA at 36.04 and ZERO at 35.69, Emo-TTA leads by over 4 points.
- Best on 10 of 12 dataset–backbone cells; the exceptions are CREMA-D under PANN-14 and TESS under HTS-AT.

Beating methods that use labelled target data, without gradients and at batch size one, is the paper's strongest claim — and the comparison is like for like on encoders.

Against audio foundation models



A strict out-of-domain comparison



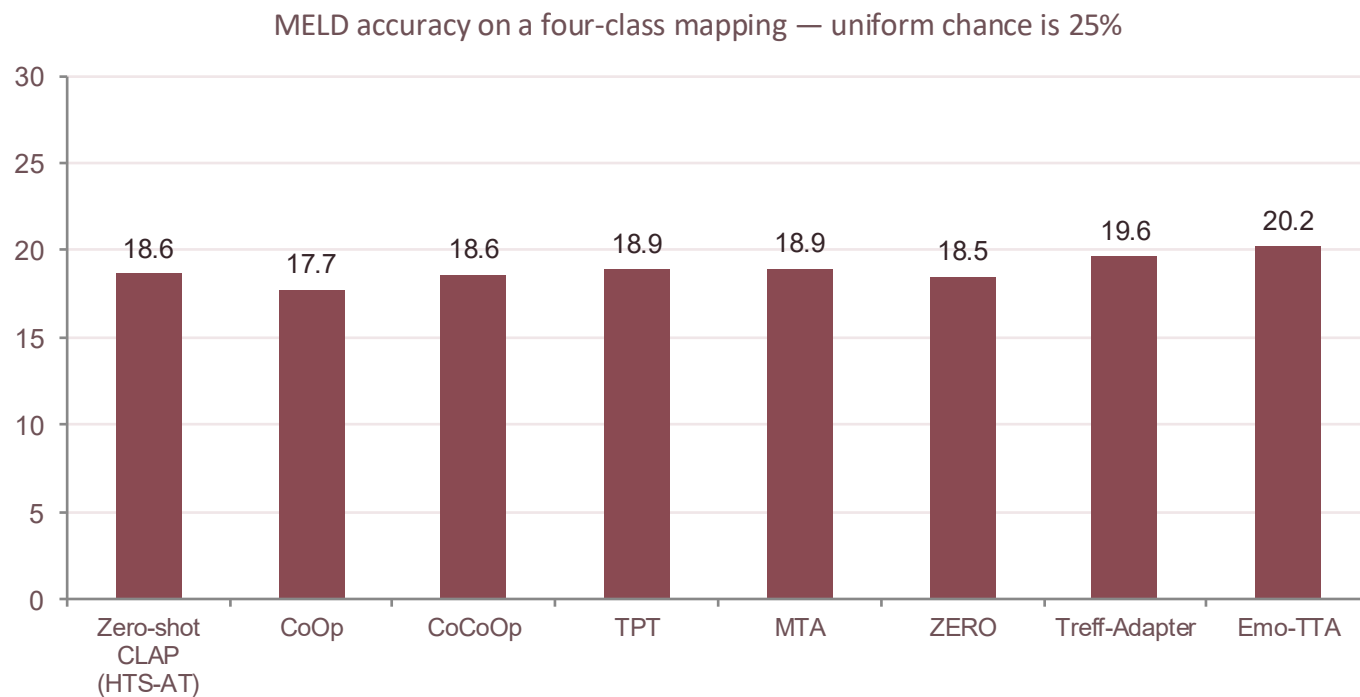
- Emo-TTA at 40.05 against the best baseline SELM at 35.77 — a gain of 4.28.
- Gains over the base CLAP model it adapts: +7.30 on IEMOCAP, +13.71 on CREMA-D, +11.86 on RAVDESS.
- Pengi appears at 30.72 — the model we read two sessions ago, now a mid-table baseline.
- The Whisper family clusters tightly at 29 to 31, which is a nice illustration that ASR-strong models are not emotion-strong.

The CREMA-D gain is the largest in the paper: 31.07 to 44.78 on a six-class task, from re-estimating a Gaussian at inference.

One number worth pausing on



MELD, mapped to four classes



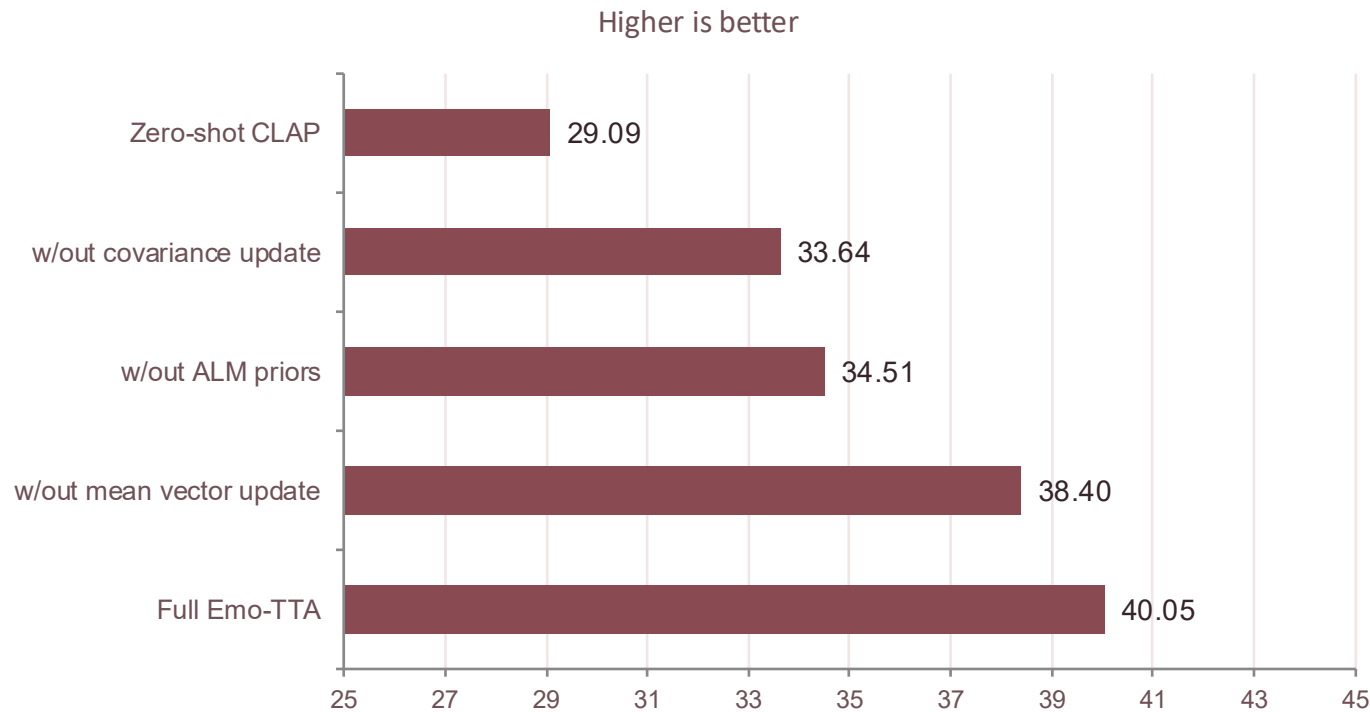
- Every method in the table scores below 25% — the accuracy a uniform random classifier would achieve on four classes.
- Emo-TTA is the best of them at 20.17, and still below chance.
- The other five corpora are comfortably above their chance levels: IEMOCAP 43.65 against 25, CREMA-D 44.78 against 16.7, RAVDESS 31.72 against 12.5.
- So MELD is not a hard case for this method — it is a case where nothing in the comparison works at all.

Improvements measured on a column where every method is below chance are hard to interpret. The honest reading is that MELD under this mapping should probably be reported separately or dropped.

Ablations



Which part of the estimation matters



- Freezing the covariance to the identity costs 6.41 points — the largest effect, and it reduces the scoring function to plain Euclidean distance.
- Removing the ALM priors — text initialisation and entropy weighting — costs 5.54.
- Freezing the class means to the CLAP text prototypes costs only 1.65.
- So the method's name notwithstanding, the biggest single contributor is learning the shape of the embedding distribution, not re-centring the classes.

That inverts the intuition the framing invites. The prototypes were always roughly in the right place; what was wrong was the metric used to measure distance to them.

4

Reading it closely

One equation, and what is not reported



The covariance update, as printed



A careful-reading exercise

$$\Sigma' = [(n_t - 1) \Sigma + \Sigma_y \gamma_{\{y,t\}} (F_t - \mu'_y)(F_t - \mu'_y)^T] / (n_t - 1)$$

● What a running average looks like

An incremental covariance estimate normally divides by the new count: the previous estimate is weighted by $n-1$ and the whole thing divided by n , so old and new evidence are averaged.

● What is printed

Numerator and denominator share the same factor, so the expression reduces to Σ plus the new term divided by $(n_t - 1)$ — an accumulation rather than an average.

● How to treat this

Most likely a typesetting slip in the denominator rather than what the implementation does — the released code would settle it in a minute. But it is worth checking, because the covariance is the component the ablation says matters most.

What is not reported



Three things the setting makes load-bearing

● Stream order

Every prediction depends on all preceding utterances, so the order of the test stream is a variable. Tent explicitly controlled for ordering by shuffling and sharing the order across methods; no equivalent control is described here.

● Hyperparameter selection

$\alpha = 0.2$ and $\beta = 4.5$ are fixed across all six corpora, which is good practice — but how they were chosen is not stated, and in a setting with no labels there is nothing to tune on honestly.

● Adaptation trajectory

Accuracy is reported at the end of the stream. How quickly the statistics converge, and whether early predictions are worse than zero-shot, is not shown — which matters for short streams and for a conversational deployment.

● Variance

Three seeds are averaged, which is more than most papers in this series, but no standard deviations are given. Several of the margins over MTA and ZERO are within a few points.

Appraisal



What the design buys, and the open threads

● Strengths

A genuinely novel combination: training-free, single-sample, and modelling the test distribution explicitly — the third of those is new in this setting.

The reframing of CLAP's zero-shot rule as Gaussian discriminant analysis is elegant and makes classical machinery available.

Beats methods that use labelled target data, with baselines re-implemented on the same encoders.

Two backbones, six corpora, three seeds — broader evidence than most four-page papers manage.

The conservative fusion with $\alpha = 0.2$ means the frozen classifier is never overridden, so catastrophic failure is structurally unlikely.

All reported arithmetic checks out against the tables.

● Open threads

Absolute accuracies are low — 40.47% average — and MELD sits below chance for every method in the comparison.

The covariance update as printed is not a standard running estimator; the code should be checked.

A full $d \times d$ covariance is estimated with no shrinkage from soft, unlabelled responsibilities.

Stream order is a free variable and no ordering control is described.

No α sweep, so how much of the gain survives giving the fitted classifier more weight is unknown.

Accuracy only — no unweighted accuracy or per-class breakdown on imbalanced corpora.

Tent and Emo-TTA side by side



Same setting, opposite machinery

	Tent (ICLR 2021)	Emo-TTA (ICASSP 2026)
What adapts	Channel-wise affine parameters γ, β	Class means, shared covariance, priors
How it adapts	Gradient descent on the entropy loss	Closed-form EM updates
Role of entropy	The objective to minimise	A confidence weight on each update
Unit of adaptation	A batch — structurally required	A single sample
State carried	Updated model parameters	Sufficient statistics of a Gaussian mixture
Failure mode	Amplifies confident errors; collapses on hard shift	Inherits confident errors, but nothing amplifies them
Safety net	None — the model is modified	$\alpha = 0.2$ keeps the frozen classifier dominant
Domain	Image corruption, digits, segmentation	Cross-corpus speech emotion
Shared premise	The model's own confidence is usable supervision	The model's own confidence is usable supervision

The highlighted row is what both papers rest on; everything above it is a different answer to how to use it.

Take-aways



Four things to carry out of the session

- 1 Adaptation need not be optimisation**

Estimation works too. A closed-form EM update gives single-sample, gradient-free adaptation where an entropy loss structurally cannot.
- 2 Confidence is better as a weight than a target**

Entropy as a loss amplifies confident errors. Entropy as a reliability weight uses the same signal without compounding the mistake.
- 3 Check the metric, not just the prototypes**

The covariance update contributes 6.41 points and the class means 1.65. Learning the shape of the space beat re-centring the classes.
- 4 Check performance against chance**

Every method on MELD scores below 25% on four classes. Improvements on a column where nothing beats chance are not interpretable.

Questions for discussion



- 1 Every method on MELD scores below the 25% uniform-chance level for four classes. What should a paper do with a benchmark column like that — report it, separate it, or drop it?
- 2 The covariance contributes 6.41 points and the class means 1.65. What is the covariance actually learning, and could you test whether it is whitening out speaker and channel variation?
- 3 Emo-TTA carries state across the stream, so prediction depends on order. What ordering would you construct to break it, and would you expect it to break?
- 4 Tent's premise rests on a rank correlation of 0.22 between entropy and loss. Does using entropy as a weight rather than a loss need a weaker version of the same assumption, or a different one entirely?
- 5 Both papers adapt without labels and both need hyperparameters. Is there an honest protocol for selecting them — and if not, how should test-time adaptation results be reported?